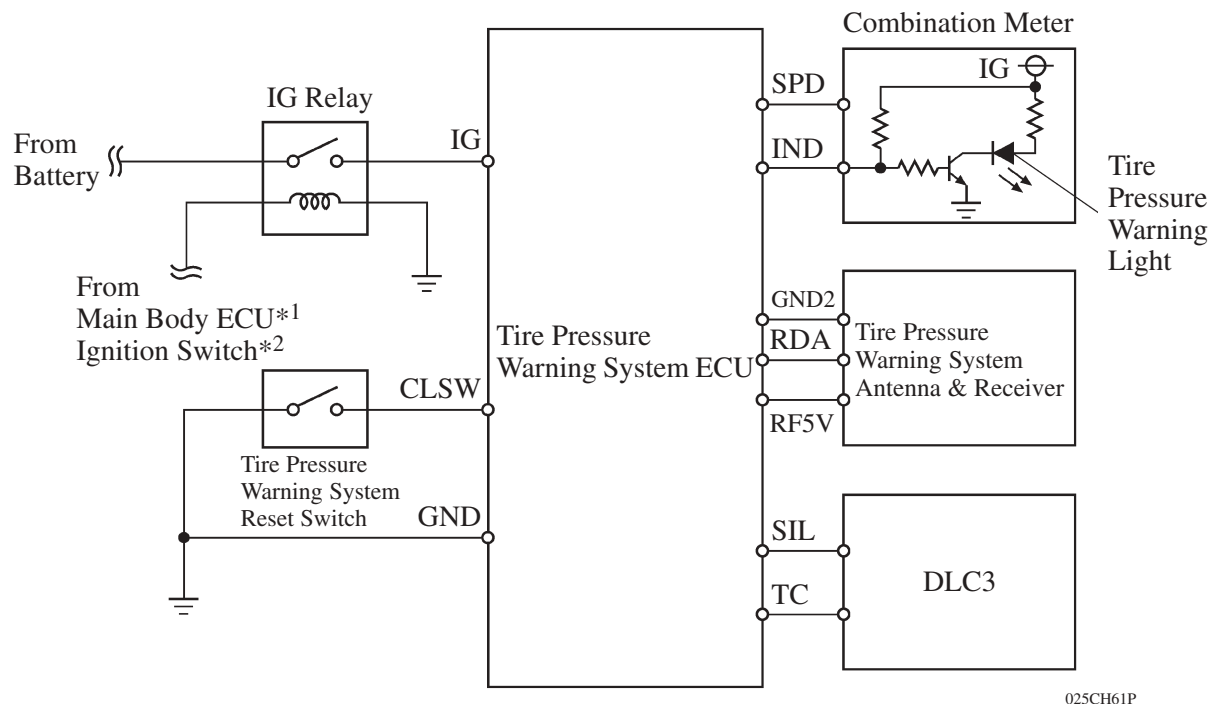


2. Tire Pressure Warning System Antenna & Receiver and Tire Pressure Warning System ECU

- The tire pressure warning system antenna & receiver, receives the radio wave signals from the tire pressure warning system valve & transmitter and transmits those signals to the tire pressure warning system ECU.
- The tire pressure warning system antenna & receiver receives the tire inflation pressure data and sensor identification data.
- The tire pressure warning system antenna & receiver can determine from the received data whether the signals came from its own tires.

► Wiring Diagram ◀



*1: Models with smart key system

*2: Except models with smart key system

2. Dual-stage SRS Airbag System

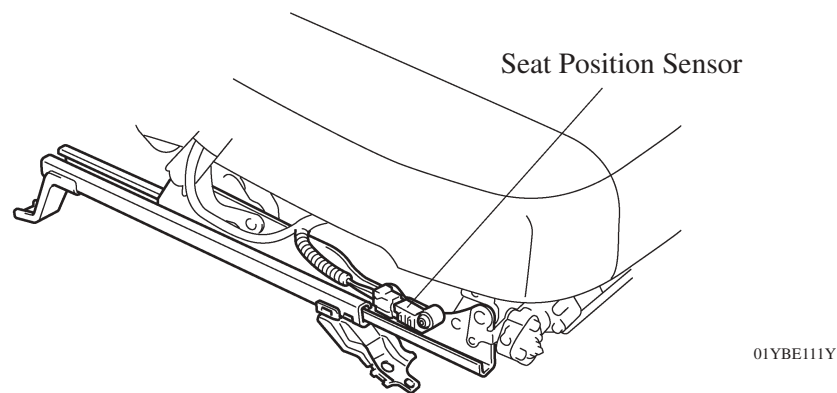
General

In this system, when the front airbag sensors and airbag sensor assembly detect a front collision, the airbag sensor assembly judges the extent of impact, seat position and whether or not the seat belts are fastened, thus optimizing the airbag inflating output by delaying the inflation timing of the 2nd initiator and the 1st initiator.

Seat Position Sensor

1) General

The seat position sensor is mounted on the upper rail portion of the driver seat rail, and includes a Hall IC and a magnet. This sensor is used to detect the sliding position of the driver seat.

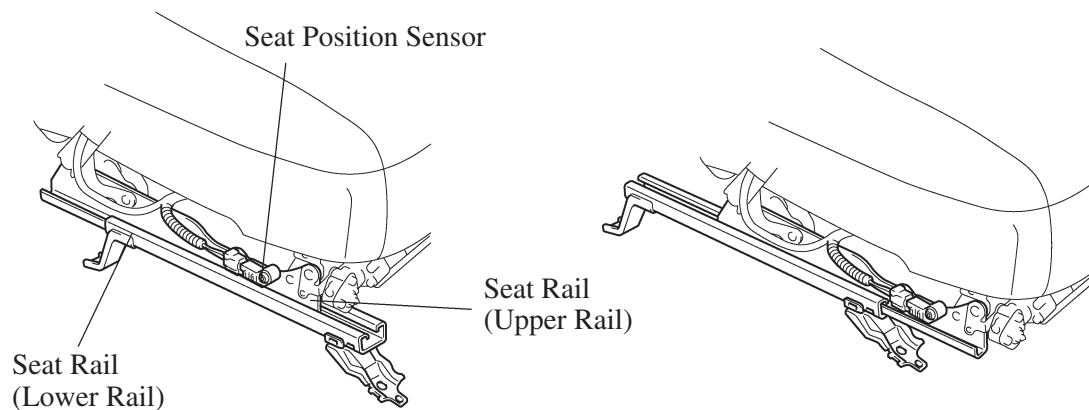


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2) Operation

When the seat is in the rearward position, the lower rail portion of the seat rail is close to the seat position sensor. When it is in the forward position, the distance between the lower rail portion and the sensor becomes larger.

Thus, the magnetic flux of the magnet inside the seat position sensor varies depending on the seat position. The Hall IC detects this variation and outputs signals to the airbag sensor assembly.



Seat position is rearward

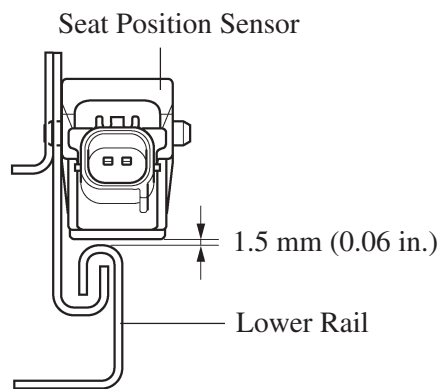
Seat position is forward

Service Tip

Follow the procedure indicated below to install the seat position sensor.

- 1) Insert a 1.5 mm (0.06 in.) feeler gauge between the seat position sensor and the lower rail portion.
- 2) Tighten the mounting bolt to the specified torque with the seat position sensor pushed down as shown.

For details, see the 2007 Camry Repair Manual (Pub. No. RM0250U).

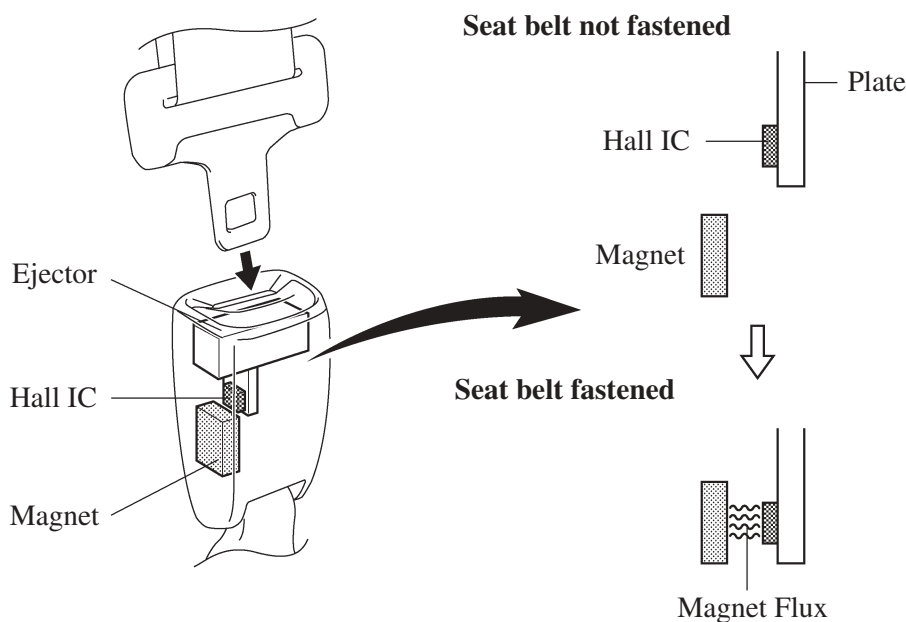


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Seat Belt Buckle Switch

The seat belt buckle switch detects whether or not the seat belt is fastened.

- The non-contact type switch is composed of a Hall IC and two magnets, installed into the front seat inner belt assembly.
- The ejector inside the front seat inner belt assembly and the plate installed to the ejector move when the seat belt is removed or inserted. The movement of the plate changes the magnetic flux density of the magnet.
- The Hall IC detects the changes in the magnetic flux density in accordance with the seat belt removal or insertion, and outputs a signal to the airbag sensor assembly (for driver seat) and occupant classification ECU (for front passenger seat).



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3. SRS Driver and Front Passenger Airbags

SRS driver and front passenger airbags contain two sets of initiators and propellants. The airbag sensor assembly helps optimize the airbag inflation speed by controlling the inflation timing of these initiators.

4. Front Airbag Sensor

Front airbag sensor uses an electrical type deceleration sensor. Based on the deceleration of the vehicle during a frontal collision, distortion is created in the sensor and converted into an electrical signal. Accordingly, the extent of the initial collision can be accurately detected.

5. SRS Knee Airbag

The knee airbag deploys simultaneously with the front airbag in a frontal collision. With the deployment of the knee airbag, the driver's lower body is restricted, thus enhancing the excellent passenger protection provided by the seat belt and front airbag.